

connector MC

connector MC	Pin connector encoder
1- White wire +5Vdc	→ 2
2- Black wire A-	→ 11
3- Black wire A+	→ 12
4- GND	→ 1
5- Black wire B-	→ 13
6- Red wire B+	→ 14
7- + Sensor temp.	(2350hm at 18°C)
8- Black wire U-	→ 3
9- Green wire U+	→ 4
10- - Sensor temp.	
11- Black wire V-	→ 5
12- Blue wire V+	→ 6
13- Not used	9, 10 (not used)
14- Black wire W-	→ 7
15- Yellow wire W+	→ 8

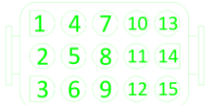
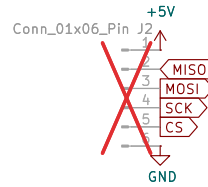


Ilustración 1. Vista superior MC / Top view MC



Ilustración 2. Vista superior encoder / Top view encoder

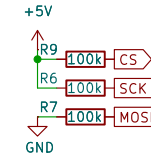
CREDIT FOR PINOUT TO ALFON <3



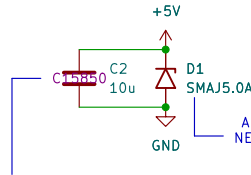
AFTER ELECTRICAL OFFSET CAL IN VX DIAG
CAN IMPROVE PRECISION A BIT BY TURNING THE MOTOR
AT RPM RANGE ABOUT 5KMH FROM DIAG READOUT WITH A DRILL
PRESSED AGAINST REAR WHEEL

NEEDS >64 ROTATIONS TO DO CAL (BUTTON HELD ABOUT 20s)
LED WILL SHOW FEEDBACK:
LED HALF BRIGHTNESS = CALIBRATION
LED DIM = CAL FAIL
LED FULL ON = CAL SUCCESS

TESTPOINTS REMOVED BECAUSE YOLO

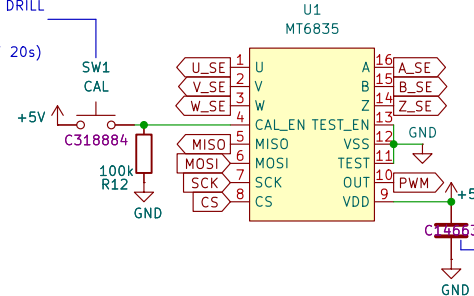


CHANGELOG:
R1
*Transceivers changed to AM26C31QDR
*TP removal
R2
*MLCC changed to soft termination
*TVS changed to 6.3V

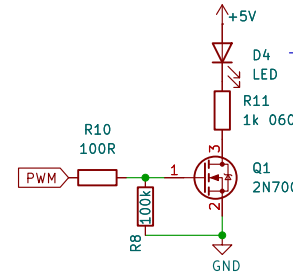
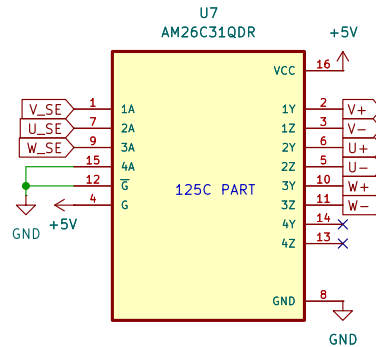
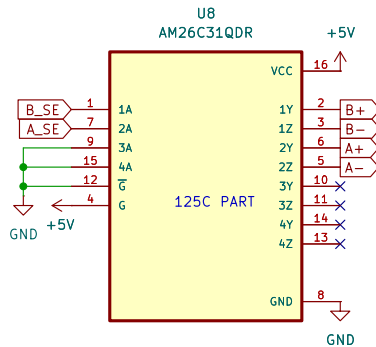


REPLACE WITH SOFT TERMINATION MLCCS
IN FUTURE!!

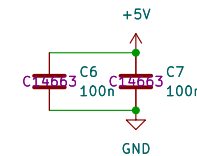
ALTHOUGH THIS TVS STARTS CLAMPING AT 6.4V
NEED TO REEVALUATE IF IT IS NOT TOO LOW FOR
OUR 5V SYSTEM.



REPLACE WITH SOFT TERMINATION MLCCS
IN FUTURE!!



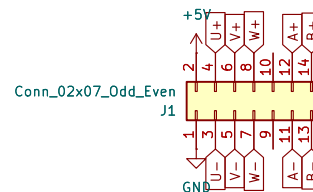
LED FOLLOWS MOTOR TURNS
(VX1 GEARBOX 5:1, SO 5 "CYCLES" PER WHEEL REV)
ALSO USED FOR USER CALIBRATION



REPLACE WITH SOFT TERMINATION MLCCS
IN FUTURE!!

0x007 = 0x08
0x008 = 0x6D
0x009 = 0x00
0x00A = 0x00
0x00B = 0x05
0x00C = 0x00
0x00D = 0x07
0x00E = 0x3C

MAKE SURE TO SET THE REGISTERS PER
VECTRIX REQUIREMENTS ONCE AND STORE IN EEPROM



LICENCE: CC-NC-SA
AUTHOR: SKRUBIS 2026 (<https://github.com/skrubis>)

Sheet: /
File: non-magic-encoder-r1.kicad_sch

Title: NON MAGIC MAGNETIC ENCODER FOR VX-1 SCOOTER

Size: A4
Date: 2026-06-18
KiCad E.D.A. 10.0.2

Rev: 1
Id: 1/1